Cable Sizing Report

AS/NZS 3008.1.1:2017 Australian conditions

Project information

calculations to see if the voltage drop across the looms for the shutdown circuit is significant or not

Date	2025-04-11
Compiled By	Jacob Lukes
Approved By	Jacob Lukes

Load

Voltage	12 V, DC
Load	1 A
Maximum voltage drop	3 %
Cable distance	6 m

Cable type

Cable type	Single-cores
Live cable	1 mm²
Neutral cable	1 mm²
Earth cable	2.5 mm²
Live conductor	Copper
Neutral conductor	Copper
Earth conductor	Copper, Flexible
Live insulation	X-110 High Temp 110°
Neutral insulation	X-110 High Temp 110°
Earth insulation	X-110 High Temp 110°

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Current rating

Rated current	20 A, Table 6, col. 11
Derated current	20 A, 20 A x 1.00
Calculated operating temperature	40°C
Maximum operating temperature	110°C

Voltage drop

Voltage drop	2.3%, 0.3 V
Voltage at load	11.7 V
Max distance	8 m for 3%
Option: Conductor temperature	Calculated
Option: Load power factor	Worst case

Impedance -line and neutral

Resistance per cable	23.3000 Ohm/km, Table 34, col. 2, 45°C
Impedance per cable	23.3000 Ohm/km

Impedance -earth

Resistance per cable	8.7600 Ohm/km, Table 37, col. 2, 45°C	0.000
Impedance per cable	8.7600 Ohm/km	

Installation

Cable installation

Wiring enclosure in air

Cable support	Bunched in an enclosure		
		Derating	Reference
Cable groups per enclosure	1	1.00	Table 22, row 2, col. 4

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Air temperature	40°C	1.00	Table 27(1), row 2, col. 7
Total derating		1.00	

Cable checks

Current rating for live and neutral conductors.	Good
Voltage drop less than 3%.	Good
Minimum earth size AS/NZS 3000.	Good

All cables

	Active Size	Earth Size	Current Rating	Volt Drop
	mm²	mm²	A	%
X	1	2.5	20	2.3
	1.5	2.5	25	1.5
	2.5	2.5	35	0.8

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